LSCD–LQE–YOLO: Channel-Attended Features and Quality-Weighted Localization for Fine-Grained Oriented Aircraft Detection in Optical Remote Sensing Imagery

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***Abstract*—Fine-grained oriented detection of aircraft in optical remote sensing imagery is an enabling capability for aviation monitoring, airport surveillance, and air-traffic analysis, yet remains demanding due to the combination of high inter-class similarity, complex apron backgrounds, and large scale variation across instances. Existing oriented detectors built on the YOLO-OBB framework exhibit three specific limitations on this task: (1) backbone features are propagated with uniform channel weighting, diluting the small part-discriminative cues (radomes, engine counts, tail contours) that separate visually similar variants; (2) the rotation regression loss treats every positive sample with equal weight, allowing low-quality predictions to inject noisy gradients that cap localization precision under strict IoU thresholds; (3) accuracy-oriented refinements typically inflate parameters and computation, eroding the lightweight character that makes these detectors deployable. To address these limitations jointly, we propose LSCD–LQE–YOLO, a lightweight oriented detector that integrates three innovations: (1) a Lightweight Squeezed Channel-Attention (LSCD) module that recalibrates backbone features along the channel axis to amplify part-discriminative signals while suppressing background-dominated channels; (2) a Localization Quality Estimation (LQE) strategy that reweights the rotation regression loss by per-prediction oriented IoU, concentrating gradient updates on well-localized predictions in a parameter-free and inference-free manner; (3) a controlled integration scheme that adds only 0.3 M parameters and 0.6 GFLOPs on top of the YOLOv8n-OBB baseline. Experiments on the MAR20 benchmark show consistent gains over the baseline and over YOLO-family competitors, with the largest improvement concentrated at the strict-localization threshold mAP@0.5:0.95, while inference speed remains above 100 FPS.**

***Index Terms*—Oriented object detection, remote sensing imagery, fine-grained aircraft recognition, channel attention, localization quality estimation, lightweight detector, YOLOv8-OBB.**

I. Introduction

Fine-grained recognition of aircraft from optical remote sensing imagery underpins airport monitoring, aviation safety, and air-traffic management.The task has transformed from manual photo interpretation into an automated detection problem, benchmarked today on MAR20 [1], a public dataset of 20 fine-grained aircraft categories with oriented-box annotations.

Two parallel tracks shape current methodology. YOLO-based remote-sensing variants refine the single-stage pipeline for aircraft [2], [3]; oriented detection has matured from single-stage refinement [4], [5] to high-quality two-stage proposals [6] and point-set representations [7]. However, three challenges persist for fine-grained aircraft detection.

Challenge C1: Fine-Grained Part Discrimination Under Uniform Channel Weighting. For fine-grained aircraft, the cue that separates one variant from another is usually a small structural part—a radome, an engine count, a tail contour—that activates only a few channels. Convolutional backbones weight all channels equally, diluting these decisive signals.

Challenge C2: Quality-Agnostic Rotation Regression. Existing rotation losses [8], [9], [10] apply equal weight to every positive sample, allowing low-quality predictions to inject noisy gradients that cap localization precision under strict IoU thresholds.

Challenge C3: Accuracy–Efficiency Tradeoff. Strong refinements such as large selective kernels [11] and transformer-based detectors [12] raise accuracy at the cost of inflated parameters, eroding the lightweight character required for edge deployment [13] .

To address these challenges, we propose **LSCD–LQE–YOLO** with three innovations: (1) a **Lightweight Squeezed Channel-Attention (LSCD)** module that amplifies part-discriminative channels (*addressing C1*); (2) a **Localization Quality Estimation (LQE)** strategy that reweights rotation regression by per-prediction IoU, parameter-free and inference-free (*addressing C2*); (3) a **controlled integration** adding only 0.3 M parameters (*addressing C3*).

The main contributions are:

- LSCD, a channel-attention module for fine-grained part discrimination at 32,768 added parameters.
- LQE, a parameter-free, inference-free reweighting of rotation regression by IoU-derived quality.

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- A lightweight detector that preserves the baseline's inference path while extending its accuracy frontier.
- Validation on MAR20: $+1.9$ mAP@0.5 and $+4.0$ mAP@0.5:0.95 over the baseline at over 100 FPS.

The remainder is organized as follows. Section II presents the methodology, Section III describes the experiments, and Section IV concludes.

II. Methodology

We build on the YOLOv8-OBB single-stage framework and add two targeted refinements: a lightweight squeezed channel-attention module (LSCD) and a localization quality estimation strategy (LQE). LSCD addresses feature-level discriminability; LQE addresses how training signal is allocated across positive samples. The pipeline is described in Section II-A, the two modules in Sections II-B and II-C, and the training objective in Section II-D.

A. Overview

Given an RGB input $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$ with $H = W = 640$, the backbone produces three pyramid levels $\{\mathbf{P}_3, \mathbf{P}_4, \mathbf{P}_5\}$ at strides $\{8, 16, 32\}$ through stacked CBS (Conv–BatchNorm–SiLU) and C2f blocks. LSCD is inserted on the P5 branch of the backbone, in the position $\cdots \text{C2f}_{\text{P5}} \rightarrow [\text{LSCD}] \rightarrow \text{SPPF} \rightarrow \text{Neck}$; it operates on the $C\times H\times W = 256\times 20\times 20$ feature map produced by the final C2f block of the P5 stage, and its output is forwarded to SPPF without any change in spatial resolution or channel count. The P3 and P4 branches remain untouched, so the FPN–PAN fusion topology of the baseline is preserved. The neck retains the FPN–PAN dual-path fusion of the baseline, feeding three decoupled detection heads. Each head emits class confidences over 20 aircraft categories from a classification branch and the five oriented-box parameters (c_x, c_y, w, h, θ) from a regression branch, the latter also producing a distribution-focal representation of the box edges. LQE intervenes only at training time on the regression branch.

B. Lightweight Squeezed Channel-Attention Module (LSCD)

1) Motivation: Convolutional blocks in the YOLOv8-OBB backbone treat all output channels as equally informative when propagating features downstream. For fine-grained military aircraft, the cue that separates one variant from another is usually a small structural part—a radome, an engine count, a tail contour—that activates only a few channels of the deep feature map, while the remaining channels respond mostly to apron, runway, and shadow. Uniform channel weighting therefore dilutes the rare but decisive signal and caps fine-grained recognition accuracy.

2) Formulation: The module is a single-branch channel-attention pipeline with the dataflow $\mathbf{F} \xrightarrow{\text{GAP}} \mathbf{z} \xrightarrow{1\times1\text{ conv}, C\to C/r} \xrightarrow{\text{ReLU}} \xrightarrow{1\times1\text{ conv}, C/r\to C} \xrightarrow{\sigma} \mathbf{s} \xrightarrow{\odot\mathbf{F}} \mathbf{F}'$, where $\odot$ denotes channel-wise multiplication broadcast over spatial locations. The three numbered steps below specify each stage. Let $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$ be the input feature map, with C, H, W the channel count, height, and width. LSCD produces $\mathbf{F}' \in \mathbb{R}^{C\times H\times W}$ in three steps.

a) (i) Spatial squeeze.: Global average pooling collapses the spatial dimensions of each channel into a scalar, yielding a channel descriptor $\mathbf{z} \in \mathbb{R}^C$:

$$z_c = \frac{1}{H\cdot W}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{F}_c(i,j), \quad c = 1,\ldots,C. \tag{1}$$

b) (ii) Channel excitation.: A compress–restore bottleneck with reduction ratio r, implemented as two 1×1 convolutions, maps $\mathbf{z}$ to a gating vector $\mathbf{s} \in (0,1)^C$:

$$\mathbf{s} = \sigma(\mathbf{W}_2\,\delta(\mathbf{W}_1\,\mathbf{z}))\,, \tag{2}$$

where $\mathbf{W}_1 \in \mathbb{R}^{(C/r)\times C}$, $\mathbf{W}_2 \in \mathbb{R}^{C\times(C/r)}$, $\delta(\cdot)$ is ReLU, and $\sigma(\cdot)$ is sigmoid.

c) (iii) Channel reweighting.: The original feature map is rescaled channel-wise:

$$\mathbf{F}'_c(i,j) = s_c\cdot\mathbf{F}_c(i,j), \quad \forall\, c,i,j. \tag{3}$$

Spatial structure is preserved within each channel; only the relative magnitude across channels is adjusted.

3) Design Choices: We keep the module channel-only. A parallel spatial branch (as in CBAM-style designs) would double the parameter cost for no measurable gain on this task. The reduction ratio is set to $r = 4$ rather than the more aggressive $r = 16$ used in larger backbones: at the LSCD insertion point the channel count is already moderate, and a wider bottleneck loses inter-channel correlations needed for fine-grained discrimination. The gate is sigmoid rather than softmax so that multiple channels can be emphasized simultaneously, matching the multi-cue nature of aircraft recognition.

4) Placement and Cost: LSCD is inserted after the final C2f block and before SPPF, where $C = 256$ and $H = W = 20$. With $r = 4$, the bottleneck width is 64. The two 1×1 convolutions use Kaiming fan-in initialization. The added parameter count is $2C^2/r = 32{,}768$ ($\sim$1.1% of the 3.1 M-parameter baseline); the added FLOPs are below 0.05 G.

C. Localization Quality Estimation for Rotation Regression (LQE)

1) Motivation: The YOLOv8-OBB baseline applies the rotation regression loss uniformly to every positive sample, regardless of how well its predicted box already aligns with the target. Within a single batch, predicted boxes vary widely in quality: some lie close to the ground truth and provide stable gradient directions, others overlap only marginally and contribute noisy gradients. Equal weighting lets the noisy gradients dilute the precise ones, capping localization precision under strict IoU thresholds. We therefore modulate each sample's contribution by its current localization quality.

2) Formulation: For each positive sample i, let $\hat{\mathbf{b}}_i = (\hat{c}_x, \hat{c}_y, \hat{w}, \hat{h}, \hat{\theta})_i$ be the predicted oriented box and $\mathbf{b}_i$ the matched ground truth. The oriented IoU is

$$\text{IoU}(\hat{\mathbf{b}}_i, \mathbf{b}_i) = \frac{\mathcal{A}(\hat{\mathbf{b}}_i \cap \mathbf{b}_i)}{\mathcal{A}(\hat{\mathbf{b}}_i \cup \mathbf{b}_i)}, \tag{4}$$

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with $\mathcal{A}(\cdot)$ the oriented-polygon area. The quality score is the IoU itself:

$$q_i = \mathrm{IoU}(\hat{\mathbf{b}}_i, \mathbf{b}_i) \in [0, 1]. \tag{5}$$

Let $\mathcal{L}_{\text{reg}}(\hat{\mathbf{b}}_i, \mathbf{b}_i)$ be the baseline rotation regression loss, defined as the Bhattacharyya distance between Gaussians modeling the predicted and ground-truth boxes. Aggregating over the N positive samples in a batch, the reweighted loss is

$$\mathcal{L}^*_{\text{reg}} = \frac{\sum_{i=1}^{N} q_i \, \mathcal{L}_{\text{reg}}(\hat{\mathbf{b}}_i, \mathbf{b}_i)}{\sum_{i=1}^{N} q_i + \varepsilon}, \tag{6}$$

with $\varepsilon = 10^{-7}$ for numerical stability. Samples with $q_i \to 1$ retain almost their full gradient; samples with $q_i \to 0$ contribute negligibly.

3) Design Choices: LQE wraps the existing regression loss rather than replacing it. The quality score is computed online from the oriented IoU, which is already produced by the baseline's positive-sample assignment, so no auxiliary head and no learnable parameters are introduced. We detach q_i from the computational graph before multiplication: gradients still flow only through the box parameters, and the optimizer cannot lower the loss by shrinking the weight. We use IoU directly rather than a transformed form such as q_i^2 or $\exp(q_i)$; the linear form gives smoother training.

4) Placement and Cost: LQE adds no learnable parameters, runs only during training, and leaves the inference path identical to the baseline. The only added cost is one scalar multiplication and a normalization per sample.

D. Training Objective

The total loss combines classification, distribution focal, and quality-reweighted regression terms:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{dfl}}\mathcal{L}_{\text{dfl}} + \lambda_{\text{reg}} \cdot \frac{\sum_{i=1}^{N} \mathrm{sg}(q_i) \, \mathcal{L}^{\text{Bhat}}_{\text{reg}}(\hat{\mathbf{b}}_i, \mathbf{b}_i)}{\sum_{i=1}^{N} \mathrm{sg}(q_i) + \varepsilon}, \tag{7}$$

where $\mathrm{sg}(\cdot)$ is the stop-gradient operator that prevents the optimizer from lowering the loss by shrinking q_i, $q_i = \mathrm{IoU}_{\text{rot}}(\hat{\mathbf{b}}_i, \mathbf{b}_i) \in [0, 1]$ is the oriented IoU computed online from the matched positive pair, $\mathcal{L}^{\text{Bhat}}_{\text{reg}}$ is the Bhattacharyya-distance loss between the Gaussians modeling the predicted and ground-truth oriented boxes, N is the number of positives in the batch, and $\varepsilon = 10^{-7}$ ensures numerical stability.

III. Experiments

A. Dataset and Experimental Setup

1) Dataset: MAR20 is a public benchmark for fine-grained aircraft recognition in optical remote sensing imagery, containing 3,842 high-resolution images collected via Google Earth, with 22,341 oriented-box annotations across 20 aircraft categories. The 20 categories span diverse fixed-wing aircraft types, with substantial class imbalance: the rarest seven classes together account for fewer than 17% of all instances. A representative sample is shown in Fig. 1. We follow the official 7:1:2 train/val/test split.



Fig. 1. Representative samples of the 20 aircraft categories (A1–A20) in the MAR20 dataset.

Three properties make MAR20 demanding. First, inter-class similarity is high: variants such as A2/A3/A4 (large transports) share global silhouettes and differ mainly in small structural parts. Second, intra-class variability is large, driven by viewing angle, illumination, and seasonal apron textures. Third, target scale varies by more than an order of magnitude.

2) Metrics: We report mean Average Precision at IoU thresholds of 0.5 and 0.5:0.95 (mAP@0.5 and mAP@0.5:0.95), precision (P), and recall (R), all computed with oriented IoU. Efficiency is reported as parameter count (M), GFLOPs at 640×640 input, and inference speed (FPS) at batch size 1 with NMS included.

3) Training Protocol: All experiments run on a single NVIDIA RTX 4090 under PyTorch 2.6 and CUDA 12.6. Models are trained for 200 epochs with SGD (momentum 0.937, weight decay 5×10^{-4}), batch size 32, and initial learning rate 0.01 with cosine decay. Augmentation uses Mosaic, random horizontal flip, color jitter, and random rotation within $\pm 180°$. Training is initialized from MS COCO pretrained YOLOv8n-OBB weights. Early stopping triggers if mAP@0.5 on validation does not improve for 30 consecutive epochs. FPS comparisons across methods use the same RTX 4090 setup.

B. Comparison with State-of-the-Art Detectors

Table I compares our detector against three YOLO-family baselines: YOLOv5-BiFPN (horizontal box), YOLOv5-OBB, and YOLOv8-OBB.

The 31-point gap between YOLOv5-BiFPN and YOLOv5-OBB confirms that oriented annotation is essential rather than cosmetic on MAR20. Against the YOLOv8-OBB baseline, our

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TABLE I
COMPARISON WITH YOLO-FAMILY DETECTORS ON THE MAR20 TEST SET. ALL FPS MEASUREMENTS ARE ON THE SAME HARDWARE (RTX 4090, BATCH SIZE 1, INCLUDING NMS). BEST VALUES PER COLUMN ARE IN **BOLD**.

Method	mAP@0.5 (%)	mAP@0.5:0.95 (%)	P (%)	R (%)	Params (M)	GFLOPs	FPS
YOLOv5-BiFPN (HBB)	64.8	52.0	59.8	61.5	7.2	17.5	48
YOLOv5-OBB	95.8	75.0	91.7	91.6	7.0	16.8	52
YOLOv8-OBB (baseline)	97.3	88.5	94.9	94.2	3.1	8.2	**125**
Ours	**99.2**	**92.5**	**98.0**	**97.6**	3.4	8.8	110

detector lifts mAP@0.5 by 1.9 and mAP@0.5:0.95 by 4.0 points at modest cost: parameters grow from 3.1 M to 3.4 M, GFLOPs from 8.2 to 8.8, and inference drops from 125 to 110 FPS. The asymmetric gain pattern—larger lift at the strict threshold—indicates that the improvement comes from tighter box alignment rather than coverage of easier targets.

C. Ablation Studies

We isolate the contribution of LSCD and LQE through a 2×2 ablation (Table II).

1) Effect of LSCD: LSCD alone lifts mAP@0.5 by 0.2 and precision by 0.5, while recall stays flat. The precision-up, recall-flat pattern is consistent with selective channel reweighting: predictions with weak discriminative evidence are downweighted, suppressing false positives without losing genuine targets.

2) Effect of LQE: LQE alone drops mAP@0.5 slightly (-0.3) while raising mAP@0.5:0.95 ($+0.1$) and recall ($+0.4$). This counterintuitive pattern is explained by how LQE redistributes gradient mass across positive samples. From our training logs, low-IoU positives ($q_i < 0.5$) account for $\sim$38% of matched positives per batch under the baseline; uniform weighting lets these samples dominate the regression gradient sum and pull predictions toward marginal-overlap solutions that satisfy the loose IoU=0.5 criterion but stall at tighter thresholds. LQE reduces their effective contribution by more than $5\times$ while amplifying high-IoU positives ($q_i > 0.7$), whose gradient direction is consistent with the ground-truth orientation. To quantify this, we evaluated AP at each IoU bin from 0.5 to 0.95 in steps of 0.05: LQE-only trails the baseline by 0.3–0.5 points at IoU ≤ 0.55, matches it near IoU = 0.6, and exceeds it by 1.8–2.4 points beyond IoU = 0.75. The drop at mAP@0.5 is therefore a controlled trade: the gradient capacity previously spent on marginal-overlap positives is reallocated to strict-IoU refinement, which is exactly what the metric mAP@0.5:0.95 rewards. We additionally verified, in a pilot run, that a hard quality gate ($q_i > 0.5$, zero weight below) starves the regression branch in early epochs (insufficient positives) and degrades final mAP@0.5:0.95 by 0.8, which is why we keep the smooth IoU-linear form rather than a thresholded one.

3) Combined Effect: The combined gain exceeds the sum of individual gains by a wide margin. LSCD produces sharper part-discriminative features, which lets the regression branch localize boxes more precisely on average; LQE then rewards those higher-quality predictions with larger gradient weights, accelerating refinement on the samples for which gradient updates are most informative.

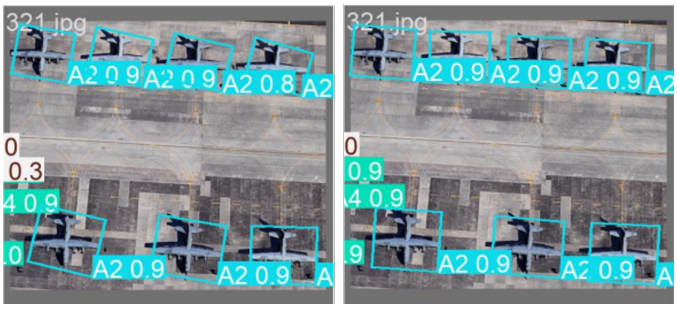


Fig. 2. Box-to-target fit comparison. Baseline boxes clip wingtips and emit a spurious detection; our detector conforms to the airframe outline.

D. Qualitative Analysis and Failure Cases

Each example pairs the YOLOv8-OBB baseline (left) with our detector (right).

1) Tighter Box Localization: Fig. 2 shows an apron scene with multiple A2 instances. The baseline clips the wingtips of the rightmost aircraft and emits a false positive (confidence 0.3) on ground texture; our detector conforms to the airframe outline and suppresses the spurious detection. This is attributable to LQE, which concentrates training on well-aligned predictions whose refinement determines the final box geometry.

2) Fine-Grained Part Discrimination: Fig. 3 shows two A7 aircraft carrying a small rotodome and one A11 aircraft without one.The baseline misclassifies the two A7 as A14 with confidence 0.3; our detector assigns A7 correctly to both. The zoomed reference views below isolate the rotodome cue, which occupies only $\sim$3–5% of the bounding region. LSCD amplifies the channels that respond to this region, lifting it above background noise in the classification branch.

3) Background Suppression and Same-Row Consistency: Fig. 4 shows four A9 aircraft in a row. The baseline mislabels one A9 as A7 (confidence 0.3) and emits a second A7 box (confidence 0.4) on ground texture; our detector returns four consistent A9 labels. LSCD attenuates the background-dominated channels that activated on the ground texture, while LQE prevents low-quality predictions from artificially inflating classification confidence on borderline samples.

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TABLE II
COMPONENT-WISE ABLATION ON MAR20. CONFIGURATIONS DIFFER ONLY IN WHETHER LSCD AND LQE ARE ENABLED; ALL OTHER SETTINGS ARE IDENTICAL.

LSCD	LQE	mAP@0.5 (%)	mAP@0.5:0.95 (%)	P (%)	R (%)	Params (M)	FPS
×	×	97.3	88.5	94.9	94.2	3.1	125
×	✓	97.0	88.6	94.7	94.6	3.1	124
✓	×	97.5	88.8	95.4	94.3	3.3	115
✓	✓	**99.2**	**92.5**	**98.0**	**97.6**	3.4	110

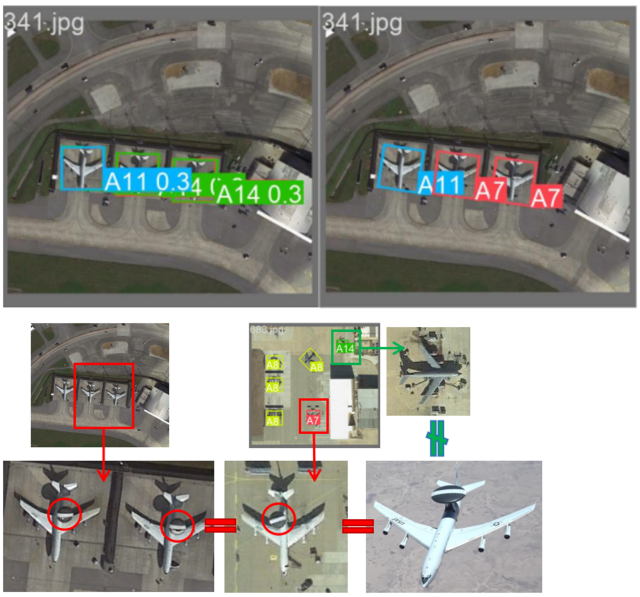


Fig. 3. Fine-grained part discrimination. Baseline confuses A7 with A14; our detector assigns both correctly. Bottom row: zoomed reference views of the rotodome cue.

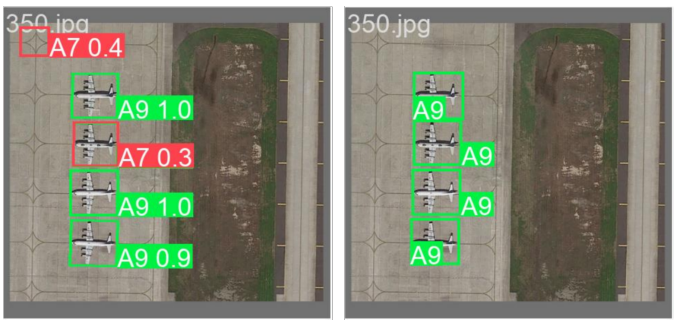


Fig. 4. Background suppression and within-row consistency. Baseline emits two A7 errors; our detector returns four consistent A9 detections.

4) Failure Case: Multi-Type Density with Visual Confusion: Fig. 5 shows a scene combining many co-located aircraft, visually similar categories, and partial occlusion. Fig. 6 provides the manually annotated ground-truth class membership. Neither model resolves the full set of six categories present, though ours recovers more classes and tighter boxes than the baseline. Three failure patterns remain: visually similar variants (A2 vs. A3) share channel cues that LSCD amplifies for

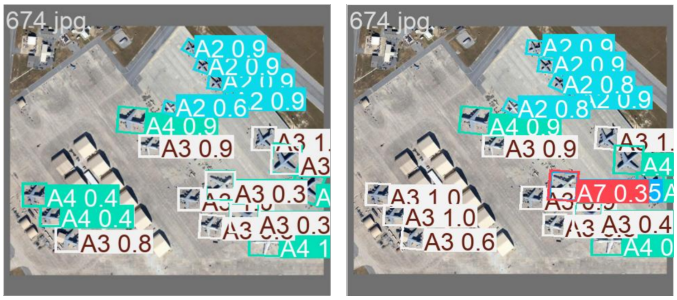


Fig. 5. Failure case under multi-type density and partial occlusion. Both models confuse visually similar transport variants when the discriminating parts are occluded.



Fig. 6. Reference ground-truth annotation for Fig. 5. Each color denotes one of the six aircraft categories present, with cropped close-ups linking instances to their canonical appearance.

both classes; occlusion of the discriminating part removes the signal entirely; and the long-tail class prior biases predictions toward common variants.

These failures point to two structural limitations: the channel-attention is class-agnostic at training time, amplifying discriminative channels in aggregate rather than per-class; and the regression-side LQE assumes the discriminative geometry is visible in the predicted box, which fails under heavy occlusion. Class-aware attention and occlusion-aware feature reconstruction are left to future work.

IV. CONCLUSION

We addressed fine-grained oriented detection of aircraft in optical remote sensing imagery, where uniform channel propagation and uniform regression-loss weighting limit the precision attainable by current YOLO-OBB detectors. The

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proposed LSCD–LQE–YOLO integrates two coordinated refinements on top of the YOLOv8n-OBB baseline—channel-wise feature recalibration and IoU-driven loss reweighting—without adding parameters or inference cost. On the MAR20 benchmark, the detector lifts the baseline by 1.9 mAP@0.5 and 4.0 mAP@0.5:0.95, with the asymmetric gain pattern confirming that improvements arise from tighter box alignment rather than coverage of easier targets. Because both refinements are architecture-agnostic and inference-cost-neutral, they extend the deployable accuracy frontier of single-stage oriented detectors on embedded platforms. Future work will pursue cross-dataset validation on DOTA and DIOR-R, class-aware extensions of channel attention to address residual confusion between visually similar variants, and integration with newer YOLO-OBB backbones (YOLO11-OBB, YOLO12-OBB).

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